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SCUOLA DI INGEGNERIA INDUSTRIALE
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AEROSERVOELASTICITY OF FIXED AND ROTARY WING AIRCRAFT – A.A. 2025-2026

Workshop 2 - Puma Blade

Alejandro Rivera Míguez

Professor:
Giuseppe Quaranta

1. Introduction

This report presents an aeroelastic stability analysis of a Puma helicopter blade in hover, specifically investigating the pitch-flap flutter phenomenon. Using the structural and inertial properties representative of the Puma rotor, the primary objective is to generate stability maps that delineate safe operational boundaries as a function of the pitch link stiffness and the chordwise position of the center of gravity. To achieve this, the study progressively advances from a simplified quasi-steady aerodynamic approach to a fully unsteady formulation.

The computational workflow relies on a modal-based Galerkin projection. The foundational uncoupled bending and torsion modes are extracted using a 1D Finite Element Method (FEM) framework provided for this course. Building upon this structural baseline, the core contribution of this work involves coupling these independent modes by assembling the global mass and stiffness matrices. This step requires the explicit calculation and integration of critical rotary-wing phenomena, most notably centrifugal stiffening and inertial coupling. The resulting coupled structural modes are subsequently mass-normalized and utilized as the basis vectors to project the distributed aerodynamic loads.

The dynamic aeroelastic stability is evaluated by constructing the generalized mass, damping, and stiffness matrices of the complete system. For the unsteady aerodynamic models, an iterative eigenvalue solver is implemented to continuously update the reduced frequency of the Theodorsen function until kinematic and aerodynamic convergence is reached.

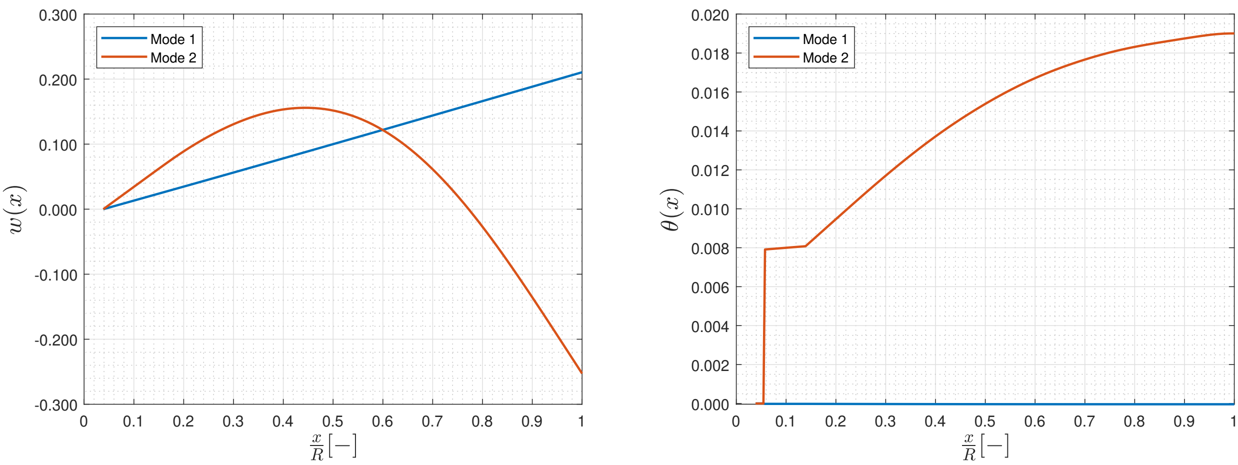
Specifically, the progressive analysis is structured around the following sequence of analytical tasks:

- **Task 1: Pitch-Flap Flutter using Quasi-Steady Aerodynamics.** Considering the fundamental 2-DOF system (first flap and first torsion modes), the quasi-steady aerodynamic matrices are assembled. The eigenvalues are computed to establish the baseline stability of the reference Puma model.
- **Task 2: Unsteady Aerodynamics at a Reference Section.** The stability of the reference model is re-evaluated using an unsteady aerodynamic formulation. The reduced frequency is computed at a single representative section located at 75% (3/4) of the blade span. The impact of the unsteady wake is quantified by evaluating the percentage change in the damping ratio compared to the quasi-steady baseline.
- **Task 3: Stability Map Generation.** A parametric sweep is conducted to construct a stability map. The pitch link stiffness is varied from 0 up to its nominal value, while the center of gravity

is shifted between 24% and 40% of the chord, identifying the critical boundary that separates stable and unstable (flutter) configurations.

- **Task 4: Multi-Mode Analysis (Optional).** The stability map generation is expanded to incorporate all available elastic modes, verifying the robustness of the stability boundary against higher-order structural dynamics.
- **Task 5: Exact Unsteady Aerodynamics (Optional).** A high-fidelity stability map is computed by evaluating the exact local reduced frequency for every section of the blade. This advanced implementation requires the aerodynamic stiffness and damping functions (K_{aer} and C_{aer}) to be evaluated continuously within the spanwise integrals.

2. Task 1: Pitch-Flap Flutter with Quasi-Steady Aerodynamics



(a) Bending shapes (b) Torsion shapes

Figure 1: First two proper orthogonal modes

The aeroelastic stability of the Puma blade in nominal hover is evaluated using its first two mass-normalized coupled modes ($\omega_1 = 27.59$ rad/s, $\omega_2 = 76.28$ rad/s). As shown in the previous figures, Mode 1 is dominated by rigid flapping, while Mode 2 exhibits strong torsion-bending coupling, accurately capturing the non-zero root rotation allowed by the pitch link compliance. Normalizing these vectors to the mass matrix ($\mathbf{M}_{gen} = \mathbf{I}$) ensures robust scaling for the aeroelastic projection.
Aerodynamic loads are modeled using a quasi-steady strip theory. Assuming the period of oscillation is much longer than the flow residence time ($k \ll 1$), Theodorsen's function reduces to unity ($C(k) = 1$). This neglects wake phase lags and simplifies the formulation into a standard linear eigenvalue problem, completely avoiding the need for iterative solvers.
Solving the state-space system yields the following complex roots:

- **Mode 1 (Flap):** $\lambda_1 = -12.86 \pm 75.19i$ 1/s $\quad \rightarrow \quad \zeta_1 = 0.6144$
- **Mode 2 (Torsion):** $\lambda_2 = -16.95 \pm 21.77i$ 1/s $\quad \rightarrow \quad \zeta_2 = 0.1686$

Since the real parts of all eigenvalues are strictly negative, the reference Puma configuration is globally stable. As expected, the flapping-dominated mode is heavily damped by aerodynamic forces, whereas the torsion-dominated mode exhibits a lower, yet sufficient, stability margin.

3. Task 2: Pitch-Flap Flutter with Unsteady Reference Section Aerodynamics

In this section, the aerodynamic model is refined by introducing unsteady effects via Theodorsen's theory. As specified in the assignment, the reduced frequency k is assumed uniform along the blade and evaluated at a reference station located at 75% of the span ($r_{ref} = 0.75R$). This approach captures circulatory wake phase lags without requiring a full spanwise integration of $k(r)$.

Because the aerodynamic matrices now depend on k (which is a function of the unknown oscillation frequency), the system cannot be solved directly. Instead, a standard **iterative $p - k$ method** is implemented. For each mode, the eigenvalue problem is solved and k is updated iteratively until the assumed aerodynamic frequency matches the imaginary part of the structural response within a strict tolerance ($\epsilon < 10^{-5}$).

The converged state-space eigenvalues for the reference configuration are:

- **Flap-dominated mode:** $\lambda = -9.96 \pm 73.35i$ [1/s] $\quad\rightarrow\quad \zeta_{torsion} = 0.5607$
- **Torsion-dominated mode:** $\lambda = -14.15 \pm 20.89i$ [1/s] $\quad\rightarrow\quad \zeta_{flap} = 0.1345$

While the system remains globally stable, the introduction of unsteady aerodynamics reveals a significant reduction in stability margins compared to the quasi-steady baseline. Specifically, the damping ratio drops by **8.74%** (from 0.6144 to 0.5607) for the flap mode, and by **20.23%** (from 0.1686 to 0.1345) for the torsion mode.

This substantial decrease demonstrates that the quasi-steady assumption is non-conservative and overestimates the damping, highlighting the absolute necessity of retaining Theodorsen's unsteady formulation for reliable pitch-flap stability predictions.

4. Task 3: Pitch-Flap Stability Map

4.1. Parametric Sweep Methodology

To fully characterize the aeroelastic safety margins of the Puma blade, a stability map was generated by performing a two-dimensional parametric sweep. The analysis explores the design space defined by two critical variables:

1. **Pitch Link Stiffness (K_θ):** Varied from 0% (completely soft) up to 100% of its nominal value.
2. **Center of Gravity Position (x_{cg}):** Swept from 24% to 40% of the local chord.

In this analysis, the chordwise shift of the center of gravity was applied uniformly along the entire span of the blade. While modifying only the outboard aerodynamic sections might represent a more localized physical modification, assuming a uniform CG shift across the entire blade provides a conservative assessment. It maximizes the inertial coupling effect between bending and torsion, isolating the fundamental physics of pitch-flap flutter.

Because the CG position directly dictates the static unbalance, the inertial coupling terms in the mass matrix change at every point of the grid. Consequently, the numerical framework recalculates the coupled structural mass matrix ($\mathbf{M}_{gen}$), re-normalizes the modes, and executes the complete unsteady $p - k$ iterative solver for each (K_θ, x_{cg}) combination.

A critical filter was implemented in the root-tracking algorithm: **any configuration yielding a null oscillation frequency ($f \approx 0$) was immediately discarded from the dynamic assessment**. A zero-frequency root indicates that the mode has lost its oscillatory nature, transitioning from a dynamic instability (flutter) to a purely static instability (such as centrifugal buckling or divergence). Furthermore, the $p - k$ method and Theodorsen's unsteady formulation inherently assume harmonic motion ($k > 0$). Thus, discarding $f = 0$ roots prevents numerical singularities and ensures that the algorithm strictly isolates the pitch-flap flutter boundary. A configuration is ultimately flagged as unstable only if a valid oscillatory mode exhibits a negative damping ratio ($\zeta < 0$).

4.2. Results and Stability Boundary

The resulting stability map is presented in Figure 2, illustrating the boundary between the safe (stable) and unsafe (flutter) operational regimes.

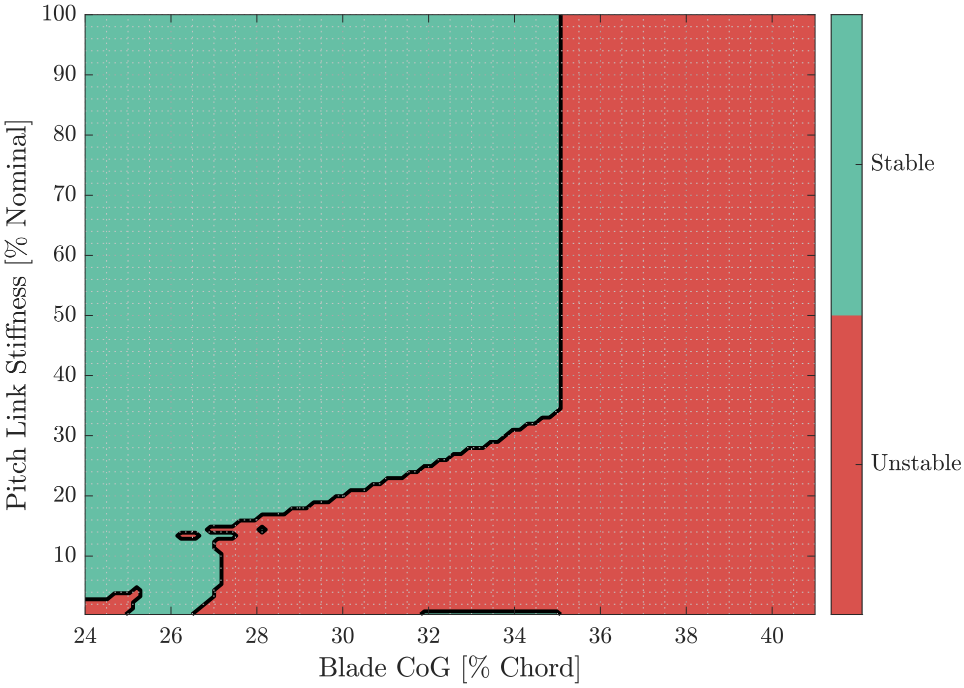


Figure 2: Stability map for the Puma blade using the first two coupled modes and unsteady reference section aerodynamics. The boundary separates stable (green/left) and unstable (red/right) configurations.

The physical trends observed in the stability map align with the theoretical expectations for rotary-wing aeroelasticity:

- **Destabilizing effect of aft CG:** Moving the center of gravity towards the trailing edge (increasing % chord) is highly destabilizing. An aft CG increases the inertial pitching moment induced by the blade's flapping accelerations. This strong inertial coupling feeds energy from the flapping motion into the torsional motion, rapidly driving the system into flutter.
- **Stabilizing effect of Pitch Link Stiffness:** A reduction in the control chain stiffness lowers the natural frequency of the torsion mode. As the torsional frequency drops and approaches the flapping frequency, modal coalescence occurs. The map clearly shows that a stiffer pitch link expands the stable region, allowing the blade to tolerate a more rearward CG position before becoming unstable.

5. Task 4: Stability Map with Enriched Modal Basis

In this section, the parametric stability analysis is extended by enriching the elastomechanical basis. Instead of relying solely on the fundamental pitch-flap coupling, the iterative unsteady $p-k$ algorithm now projects the aeroelastic system onto the first six proper orthogonal modes of the blade.
The resulting stability map for the 6-mode basis is shown in Figure 3, while a direct boundary comparison between the 2-mode and 6-mode models is presented in Figure 4.

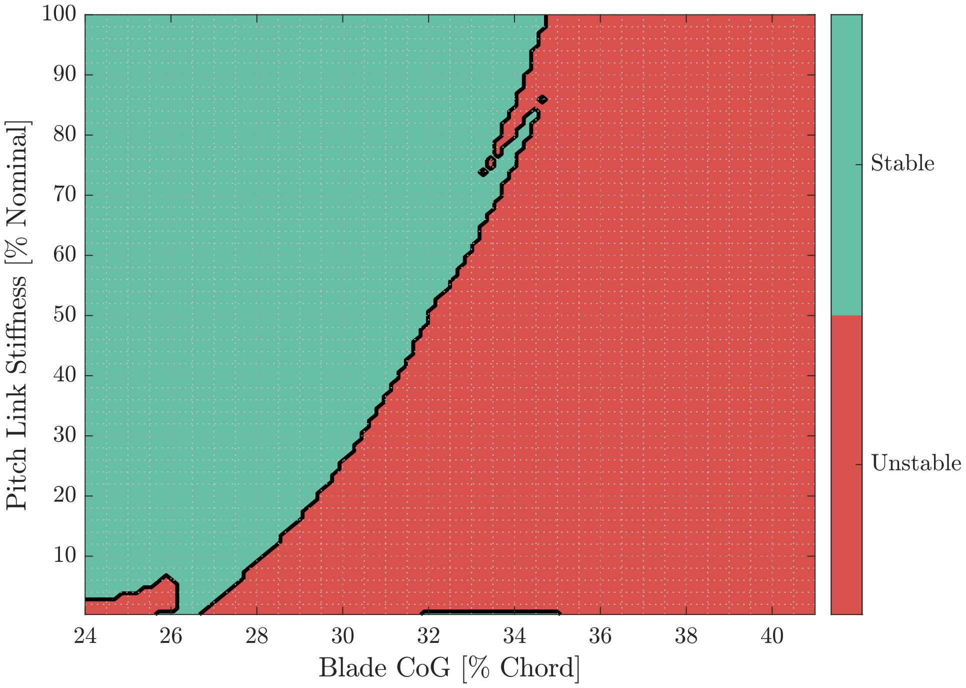


Figure 3: Stability map computed using the first six elastomechanical modes with unsteady reference section aerodynamics.

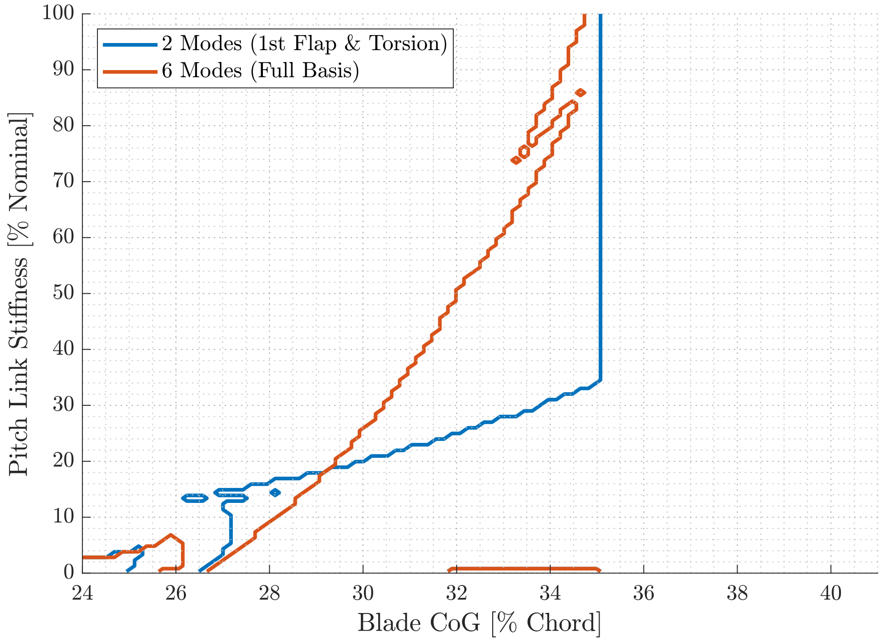


Figure 4: Comparison of the stability boundaries: 2-mode basis (blue) vs. 6-mode basis (orange).

Observing the comparative plot, it is immediately apparent that the inclusion of higher-order modes drastically alters the stability boundary, introducing new dynamic coupling mechanisms. The most critical observations are:

- **Non-conservative region (Low to Mid Stiffness):** For pitch link stiffness values below approximately 35% of the nominal value, the 6-mode boundary (orange) lies significantly to the left of the 2-mode boundary (blue). This indicates that the simplified 2-mode analysis is *non-conservative* in this regime, failing to predict flutter triggered by the interaction of higher-order bending or lag modes.
- **Conservative region:** With the CG further forward at approximately 28%, the 2-mode model remains *conservative* and imposes tighter constraints on pitch link stiffness.

In conclusion, while a 2-mode approximation successfully captures the fundamental pitch-flap physics, it is insufficient for a reliable design envelope. The severe shifts in the stability boundary highlight the absolute necessity of performing a convergence study on the modal basis before drawing definitive aeroelastic conclusions.

6. Task 5: Stability Map with Exact Unsteady Aerodynamics

In this final phase of the project, the aeroelastic framework is upgraded to eliminate the reference section assumption. Instead of using a uniform reduced frequency evaluated at 75% of the span (k_{ref}), the exact unsteady aerodynamic formulation treats the reduced frequency as a radially varying function, $k(r)$. Consequently, the unsteady aerodynamic matrices must be integrated exactly along the span at every step of the $p-k$ iterative loop.

The stability map using the 6-mode basis and exact unsteady aerodynamics is shown in Figure 5. To evaluate the impact of this aerodynamic refinement, a final comparison between all three models (2-mode reference, 6-mode reference, and 6-mode exact) is presented in Figure 6.

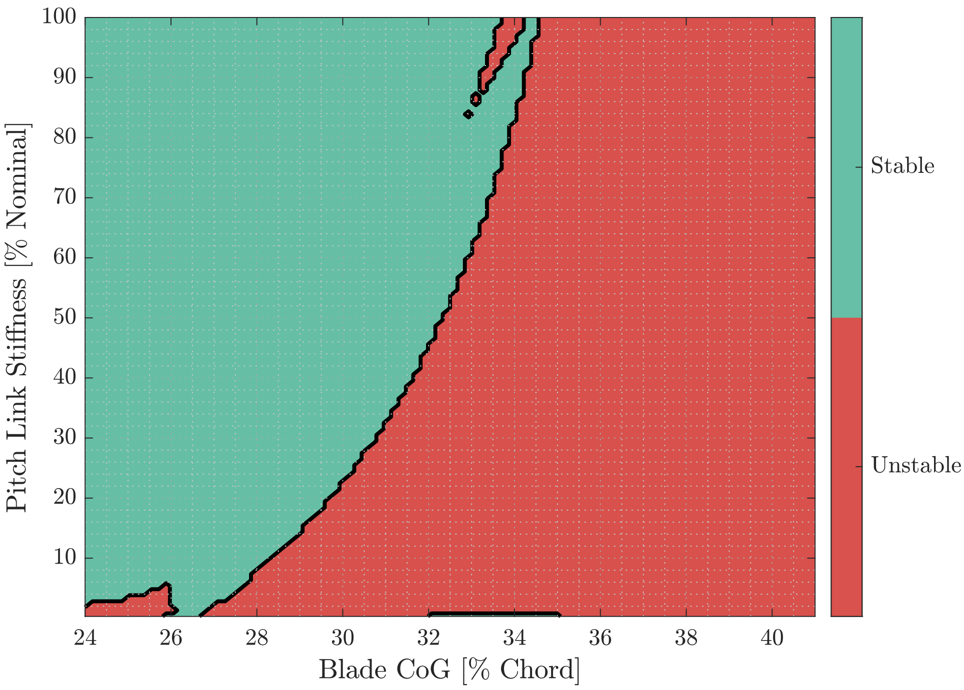


Figure 5: Stability map computed using the 6-mode basis and exact spanwise-varying unsteady aerodynamics.

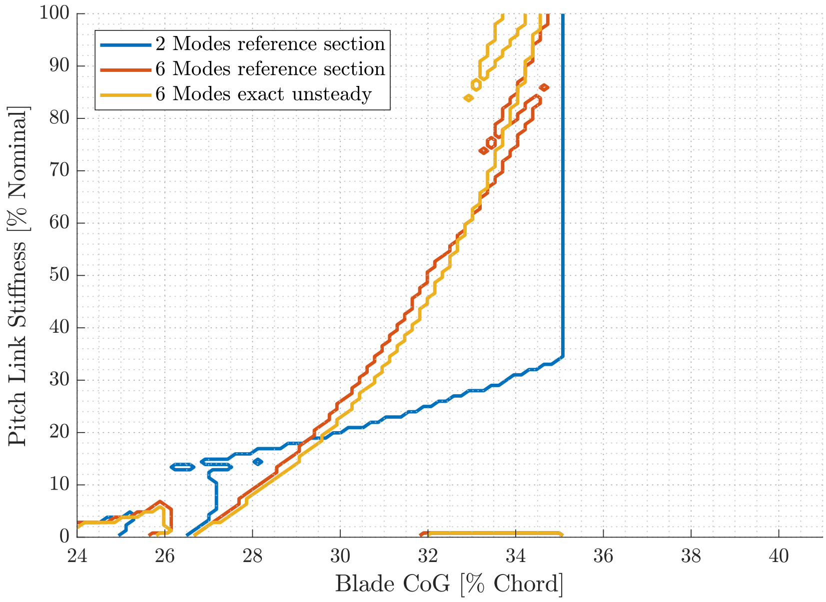


Figure 6: Global boundary comparison: 2-mode reference section (blue), 6-mode reference section (orange), and 6-mode exact unsteady (yellow).

Analyzing the final comparison plot yields several critical conclusions regarding the modeling of rotor blade aeroelasticity:

- **Validity of the Reference Section Assumption:** The stability boundary for the exact unsteady model (yellow line) tracks the reference section model (orange line) remarkably well. Both curves share the exact same topological features, including the high-frequency modal coalescence notch around 80% stiffness. The discrepancies between the two are minor, proving that the $r_{ref} = 0.75R$ assumption is a highly accurate approximation for this rotorcraft analysis.
- **Computational Efficiency:** Evaluating the exact unsteady aerodynamics requires computing complex radial integrals inside every iteration of the $p - k$ solver, drastically increasing the runtime. Given the minimal shift in the stability boundary, the reference section approach is proven to be an optimal engineering trade-off, delivering near-exact results at a fraction of the computational cost.
- **Structural vs. Aerodynamic Refinement:** The comparison clearly demonstrates that the stability map is far more sensitive to the *structural* discretization (moving from 2 modes to 6 modes) than to the *aerodynamic* refinement (moving from reference section to exact unsteady).

7. Conclusions

This aeroelastic analysis of the Puma helicopter blade yields the following key takeaways:

- **Stability Drivers:** Moving the center of gravity aft strongly destabilizes the blade due to inertial pitch-flap coupling. Conversely, increasing the pitch link stiffness prevents modal coalescence and expands the safe operational envelope.
- **Structural Discretization:** A simple 2-mode approximation is non-conservative. Expanding to a 6-mode basis is strictly necessary to capture higher-order modal interactions that significantly alter the stability boundary.
- **Aerodynamic Modeling:** Unsteady aerodynamics must be included to avoid overestimating damping. However, evaluating the reduced frequency at a 75% reference section provides near-exact results compared to a full spanwise integration, saving massive computational time.

In conclusion, for preliminary design and broad parametric sweeps, employing a 6-mode structural basis coupled with 75% reference section unsteady aerodynamics offers the best balance of accuracy and computational performance.